

Exercise 5

Define disjunctive normal form formulas. When would it be helpful to have a formula in DNF? Sketch in pseudocode an algorithm to turn any propositional formula into a DNF formula.

Part 1: Define disjunctive normal form formulas.

Answer

A formula is in disjunctive normal form (DNF) if it is a finite disjunction of conjunctions of *literals*, where a literal is either an atomic proposition p or its negation $\neg p$. Formally:

$$\varphi \text{ is DNF} \iff \varphi \equiv \bigvee_{i=1}^k \bigwedge_{j=1}^{m_i} \ell_{ij},$$

with each $\ell_{ij} \in \{p, \neg p\}$. By convention, $\top$ is the empty conjunction and $\perp$ is the empty disjunction.

Part 2: When would it be helpful to have a formula in DNF?

Answer

Disjunctive normal form formulas are useful to evaluate satisfiability and represent logical circuits.

Part 3: Sketch in pseudocode an algorithm to turn any propositional formula into a DNF formula.

Answer

Algorithm 1 DNF(F)

$F \leftarrow \text{REMOVE_IMPLICATIONS}(F)$	$\triangleright$ remove $\rightarrow, \leftrightarrow$
$F \leftarrow \text{PUSH_NEGATION}(F)$	$\triangleright$ move $\neg$ inward (De Morgan)
$F \leftarrow \text{DISTRIBUTE_OR}(F)$	$\triangleright$ distribute $\vee$ over $\wedge$
return SIMPLIFIED(F)	
